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Global Warming and Neurological Practice: Systematic Review

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ABSTRACT

Background: Climate change, including global warming, is expected to cause poorer global health and a rise in the number of environmental refugees. As neurological disorders account for a major share of worldwide morbidity and mortality, climate change and global warming are also destined to alter neurological practice; however, to what extent and by which mechanisms is unknown. We aimed to collect the available information on the effects of ambient temperatures and human migration on the epidemiological and clinical manifestations of major neurological disorders.

Methods: We searched PubMed and Scopus from January 1, 2000 to November 30, 2020 for human studies published in English addressing the influence of ambient temperatures and human migration on Alzheimer's and non-Alzheimer's dementia, epilepsy, headache and migraine, multiple sclerosis, Parkinson's disease, stroke, and tick-borne encephalitis (as a model disease for neuroinfections). The protocol was pre-registered at PROSPERO (2020 CRD42020147543).

Results: 101 studies met inclusion criteria, but we were unable to identify a single study addressing how global warming and human migration will change neurological practice. Still, extracted data suggested multiple ways by which these aspects might alter neurological morbidity and mortality in the future.

Conclusion: Significant heterogeneity exists across studies with respect to methodology, outcome measures, control of confounders and study design, but there is enough evidence to suggest climate change will affect the neurological practice of all major neurological disorders. Adequately designed studies to address this issue are urgently needed, which will require concerted efforts from the neurological community.

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INTRODUCTION

The United Nations have identified climate change, which includes global warming, as the "defining issue of our time"[1]. As stated in the Paris Climate Agreement, the increase of global temperature must be contained within 2, ideally 1.5, degrees Celsius above pre-industrial levels[2] to avoid major negative consequences, including poorer global health and a rise of environmental refugees (i.e. displaced people from increasingly inhabitable regions of the world). Thus, climate change and global warming will likely worsen illnesses worldwide[3] and may become the major drivers of human migration[4].

According to the Global Burden of Disease Study, neurological disorders are the foremost cause of disability adjusted life years (DALYs), accounting for 10.2 percent of global DALYs, and neurological diseases are the second-leading cause of deaths, representing 16.8 percent of global deaths[5]. It is therefore reasonable to assume that climate change and global warming will also have a major impact on clinical neurological practice.

To start addressing this issue, one must first identify the areas of neurological practice that will likely be subject to alterations related to global warming. It is necessary to investigate in which ways a rise of ambient temperatures might affect the frequency, semiology, and outcome of major neurological disorders. Furthermore, the prevalence and incidence of neurological disorders in refugees and other human migrant populations might serve as a proxy for what could be expected in future environmental refugees.

In this review, our main objective was to identify how ambient temperatures influence the epidemiology and clinical aspects of major groups of neurological disorders. The second objective was to investigate the prevalence and incidence of neurological disorders in human refugee and migrant populations.

METHODS

Using the PICO approach[6], we phrased the following research questions:

1. In people with major neurological disorders, including Alzheimer's and non-Alzheimer's dementia, epilepsy, headache and migraine, multiple sclerosis, Parkinson's disease, stroke, and tick-borne encephalitis (as a proxy for neuroinfectious diseases) (Population), how does an increase in ambient temperatures (Intervention), as compared to normal ambient temperatures (Comparison), affect the frequency, symptomatology, and mortality of these disorders (Outcome)?
2. In human refugee and migrant populations (Population), how does the fact of being displaced (Intervention), as compared to populations living in their home country (Comparison), affect the prevalence and incidence of neurological disorders (Outcome)?

For each PICO question a systematic review of the literature was performed using a predefined search. The full search strategy (including MeSH headings for searches in PubMed) is available from the ***Supplemental file 1***. The review was pre-registered at PROSPERO (2020 CRD42020147543).

Types of studies

We evaluated all cross-sectional or longitudinal, retrospective or prospective, observational clinical and research studies on major neurological diseases with epidemiological and clinical data associated with ambient

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temperatures, as well as epidemiological and clinical data associated with human and refugee populations. We excluded interviews, editorials, opinions, and news articles.

Electronic search strategy

We included human only studies published in English and listed in PubMed and Scopus from January 1st, 2000 to November 30, 2020. Non-English literature was included if an English abstract and a reliable translation of the manuscript into English were available. The literature search was supervised by an information specialist from the University of Copenhagen's library service, and the search strategy was developed in accordance with the above PICO questions. For search examples and filters, the reader is referred to the ***Supplemental file 1***. The references of relevant articles were manually searched to identify additional articles. Further, papers were cross-referenced using the 'cited by' function in PubMed. When necessary, personal communication with authors was attempted by email or phone to obtain additional data.

Data collection

After reviewing titles and abstracts, relevant studies were assessed on a full text basis. Data was extracted by the 1st, 2nd and 3rd authors and cross-checked by the senior author; any uncertainties concerning data extraction and interpretation were resolved by consensus.

Statistical analyses

Owing to the high heterogeneity of the data including different definitions of temperature thresholds, quantitative statistical analysis was not performed.

RESULTS

The primary search yielded 4,992 titles. After screening and removal of duplicates, 101 studies met the inclusion criteria for the final review (**Figure 1**). **Table 1** provides an overview of studies (n=85) regarding ambient temperatures and major neurological disorders. The effects of migration on prevalence and incidence of neurological disorders (n=16 studies) are outlined in **Tables 2** and **3**. Extracted raw data from the literature search are available on request from the corresponding author.

Ambient temperature and major neurological disorders

Alzheimer's and non-Alzheimer's dementia

Ten studies investigated the effects of high ambient temperatures on patients with Alzheimer's and non-Alzheimer's dementia[7-16]. The studies were from 7 countries (AU, CN, ES, IT, KR, UK, US, and VN) with a total case population of N = 960,675. One study did not specify the number of cases[8].

Maximum temperatures reported were >30°C in 8 studies[7, 8,10-13,15-17] and 18°C, respectively, >24.5°C in two other studies[9,14]. In 8 of the 10 studies, high ambient temperatures were associated with worsening of symptoms including agitation, as well as an increase in hospitalizations and/or mortality[7,8,10,12-17]. Five studies suggested that hot temperature were attributable to worsening of symptoms and increased risk of admission[7,12,14-16], while Hansen et al. found a significantly increased rate of both hospital admissions and

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mortality in 94,447 patients with dementia during periods of heat waves[8]. Zanobetti et al, Xu et al. and Conti et al. also found significantly increased mortality during hot months[10,13,17]. In the remaining two studies, no associations were found between high ambient temperatures and hospital admissions or death in people with dementia [9,11]. Overall, increased temperatures were associated with increased rates of hospitalization and mortality in patients with dementia in 8 of the 10 studies.

Epilepsy

Only one epilepsy study met inclusion criteria, a case-crossover study with 604 patients[18]. The results indicated a 46% lower risk of admissions for epileptic seizures one day after exposure to temperatures above 20°C. Further, an inverse association was found between epileptic seizures and low atmospheric pressure and high humidity[18].

Headache and migraine

Two studies from 2 countries (US and FR) including N = 7,156 cases reported on the effects of high temperatures on headaches, including 33% migraine [19,20]. The largest study by Mukamal et al. analyzed data from 7,054 patients seen in an emergency department between 2000-2007[19]. In this study, higher ambient temperatures in the 24 hours preceding hospital presentation increased the risk of acute headache requiring emergency evaluation with 7.5% for each 5°C increment in temperature. Neut et al. interviewed 102 children and adolescents with migraine and/or their parents about triggering factors precipitating migraine attacks. Seventy % stated warm climate could trigger their migraine, and 24% reported warm climate was often or very often a trigger factor for migraine attacks[20].

Multiple sclerosis

Six studies from 5 countries (US, FR, DE, AU, JP) with a total N = 5,305 addressed the effects of high temperatures on patients with multiple sclerosis [21-26]. Four studies found high temperatures to be associated with worsening of symptoms in these patients [21,22,24,26]. Stellmann et al. included 1,254 patients and showed that deficits measured by a walking test slightly worsened when temperatures increased to 20°C[26]. In an anonymized survey by Simmons et al., 70% of multiple sclerosis participants reported worsening of their symptoms with high temperatures[22]. Poor cognitive performance significantly correlated with warmer outdoor temperatures in another group of 40 multiple sclerosis patients[24]. Furthermore, the latter study also found a significant correlation between poor cognitive performance and high temperatures over a period of 6 months, suggesting a negative impact of warm temperatures on cognitive function in multiple sclerosis patients when analyzed cross-sectionally and longitudinally. Higher frequencies of multiple sclerosis attacks were recorded during the warmest months in a small sample of 34 multiple sclerosis patients[21]. The remaining two studies did not show any effects of high temperatures on multiple sclerosis symptomatology [23,25]. Thus, 4 of 6 studies reported worsening of symptoms in multiple sclerosis patients during warm periods.

Parkinson's disease

Three studies from 3 countries (FR, ES and US) with a total N = 204,656 evaluated the effects of high temperatures on Parkinson's disease [10,27,28]. Results from one study indicated a correlation between high

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ambient temperature (>34°C) and an increase risk of excess morbidity and mortality in Parkinson patients[28]. In a small French study with 36 Parkinson's disease patients, there was a trend towards more frequent symptoms of autonomic failure during heat wave periods[27]. In contrast, another study investigated the effects of extreme hot days (maximum temperature of 31,7°C) on mortality in 201,333 patients with a diagnosis of Parkinson's disease but found no association between mortality and ambient temperatures[10].

Stroke, overview

In total, 59 papers dealing with the effects of ambient temperatures on cerebrovascular morbidity and mortality were included [10,29-60,e61-67]. The studies were from 18 countries (AU, CA, CN, DE, DK, ES, GB, IN, IL, IT, JP, KR, PR, QA, RU, SE, TU, TW, and US) with a total case population of N = 5,869,284. Ischemic stroke was reported in 1,812,457, hemorrhagic stroke in 446,407, and SAH in 7,160 cases. In the remaining 3,603,260 cases, stroke subtypes were not specified.

Stroke, hospitalizations

Associations between stroke-related hospitalization and ambient temperatures were reported in 38 studies. Thirteen studies reported increasing risks for hospitalizations for stroke with higher ambient temperatures [29, 36, 44,46,50,56,e65-71]. Maximum temperatures were 12.9°C in one study[29], 20-27°C in 5 studies[36,44,e65,69,70], and >30°C in 5 studies[47,50,56,e66,71]. Temperatures were unspecified in two studies[e67, 68]. Chen et al. found excess hospitalizations due to ischemic stroke with high temperatures, while the risk for admissions because of hemorrhagic stroke was increased during both high and low temperatures[49]. Green et al. and Basu et al. also found increased risks of hospitalization due to ischemic stroke with high temperatures (25.3 and 30.1°C, respectively), but a decreased risk for hemorrhagic stroke[37, 42]. Bai et al. reported an increase in risk of stroke admissions both with high and low ambient temperatures[e62]. In 10 studies, lower ambient temperatures (-16.94 to 17 °C) were associated with increased risk of stroke-related hospitalizations[53,57,e64,72-78]. In the study of Wang et al., there was an inverse correlation between temperatures and ischemic stroke admissions, leading to an increase of admissions during cold spells and a decrease during heat waves[48]. Lin et al. reported lower risks of stroke admissions associated with high ambient temperatures (31.7°C), even though the results were not statistically significant[35]. Cevik et al. only found association between lower temperature and increased risk of SAH[e79], and like the remaining eight studies no associations between ambient temperatures and stroke-related hospitalizations was found[30,34,41,43,47,59,e80,81]. Overall, most studies (17 of 24) reported higher risks of stroke-related hospitalizations during periods of high ambient temperatures.

Stroke, mortality

21 studies reported on the association between ambient temperatures and cerebrovascular mortality[10,31-33,38-40,45,51,52,54,55,58,e60,61,63,77,82-85]. In 11[10,31,32,38,39,45,55,e61,82-84] of the 21 studies, increased risks of cerebrovascular death was associated with high ambient temperatures, including very high temperatures (>30°C)[10,31,32,38,39,55,e61,83,84]. Five studies reported an increase in cerebrovascular mortality with both low and high temperatures[51,52,58,60,e63], while 2 studies found no effect[33,40]. Zhang et al., Myint et al. and Yang et al. were the only to report significantly higher risks of cerebrovascular death in periods with cold, but not warm, temperatures[54,e77,85]. Overall, most

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studies (16 of the 21 studies) found a higher risk of cerebrovascular mortality with increasing ambient temperatures.

Tick-borne encephalitis

The effects of high ambient temperatures on tick-borne encephalitis were assessed in 4 studies from 3 countries (SE, CZ, SI)[e86-89]. Lindgren et al. linked an increase in the incidence of tick-borne encephalitis in Sweden, starting in the mid-80s, to climate change with increasingly milder winters and earlier springs[e87]. Lukan et al. and Zeman et al. analyzed 1,786, respectively, 8,700 cases of tick-borne encephalitis in the period of 1961-70 to 2004[e88, 89]. During 1980 to 2004 they found an increase in tick-borne encephalitis foci in areas of increasing altitudes which corresponded to gradual rises in annual temperatures, indicating an effect of climate change and warmer temperatures on local tick-borne encephalitis incidence. Overall, the studies indicated a rise in the total incidence of tick-borne encephalitis with milder temperatures in Sweden and increasing temperatures in higher altitude areas in the Czech Republic and Slovenia.

Neurological disorders in human migrant and refugee populations

Results from studies comparing migrants with populations from their country of origin are listed below and in **Table 2**.

Alzheimer's and non-Alzheimer's dementia

One observational study reported a higher prevalence of dementia in a group of elderly Japanese people (>70 years) who had migrated from Okinawa, Japan, to Brazil, as opposed to people who had stayed in Japan[e90].

Multiple Sclerosis

Four observational studies investigated the effects of migration on the frequency[e91,92] and morbidity[e93,94] of multiple sclerosis. In the study by Guimond et al., multiple sclerosis prevalence was higher in migrants who came from Iran to British Columbia[91], whereas in the study by Hammond et al. the incidence was higher in the population from the home country (UK and Ireland) compared to the incidence in a migrant population settling in Australia[e92]. Further, Merle et al. investigated visual impairment in multiple sclerosis patients from Martinique who had or had not migrated to metropolitan France for at least 1 year before the age of 15 years and found that symptoms were more frequent and severe in the non-migrant group[e93]. Comini-Frota et al. found no difference of multiple sclerosis morbidity between an Italian migrant population in Brazil and non-migrant Italians[e94].

Stroke

Three observational studies compared data on cerebrovascular disease in migrant populations with data from non-migrant populations[e95-97]. Two studies reported a higher prevalence of stroke in non-migrant groups living in China[e96] and Ghana[e95], as opposed to migrant populations living in Hong Kong, Singapore, Taiwan, Western Europe, and North America. After adjusting for conventional cardiovascular risk factors, results were no longer significant in one study[e95]. In contrast, Wolfe et al. observed a higher incidence of stroke in migrants from

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Barbados moving to South London, including an increased incidence for specific stroke subtypes such as total anterior cerebral infarctions, posterior cerebral infarctions and subarachnoid hemorrhages[e97].

Results from studies comparing migrants with populations from their country of arrival are summarized in **Table 3**.

DISCUSSION

Although the United Nations have declared climate change the greatest challenge of the 21st century and although neurological disorders comprise the greatest share of DALYs globally, we were unable to identify a single adequately designed study addressing how climate change and its consequences will alter neurological practice in the future. This is concerning given that the evidence that we did find indeed suggests many ways by which the effects of global warming and human migration on neurological practice may unfold.

Ambient temperatures and neurological disorders

Most stroke studies showed a relationship between increasing ambient temperatures and higher rates of hospitalization and mortality. Heat exposure may be associated with hemoconcentration and hyperviscosity, impaired endothelial function and hemodynamic disturbances, including cardiac arrhythmia, thereby increasing risks for both ischemic and hemorrhagic stroke[e98].

Likewise, warmer temperatures were reported to worsen cognitive symptoms and to increase mortality in dementia patients. The pathogenesis is unclear, but impaired physiological adjustment to rising temperatures because of dysautonomia are known from patients with fronto-temporal dementia and Alzheimer's disease and may cause dehydration, cardiorespiratory distress and susceptibility to drug side effects[e99]. Autonomic failure aggravated by high temperatures is also a plausible complication in Parkinson's disease that may lead to orthostatic hypotension and ensuing risk for trauma from syncope and falls[27].

With multiple sclerosis, most studies revealed a decline in motor and cognitive functions following a raise in ambient temperatures. While this is consistent with Uhthoff's phenomenon, i.e. the well-known reversible aggravation of multiple sclerosis symptoms caused by the blocking or slowing of nerve conduction with heat, no studies have specifically addressed the importance of Uhthoff's phenomenon relative to attack rates, inflammatory mechanisms and secondary neurodegeneration with rising temperatures. Furthermore, hot temperatures are trigger factors for headache and migraine[19,20]. Heat is known to cause vasodilation which may contribute to vascular migraine[e100, 101], and water deprivation can provoke secondary headaches including migraine[e102].

As to epilepsy, there are well-established relationships with temperature and pediatric febrile seizures, seizures in conjunction with heat strokes[e103] and Dravet syndrome-related seizures associated with hot water; and increased body temperature may trigger hippocampal neuronal activity in mesial temporal lobe epilepsy[e104]. In addition, using tick-borne encephalitis as a model disease for neuroinfections, we found evidence that vector-borne diseases are prone to spread from endemic areas to currently non-endemic regions with increasing humidity and rising temperatures.

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Human migration and neurological disorders

The impact of migration on the prevalence, incidence and severity of major neurological disorders is substantial but not uniform. Social, economic and cultural characteristics of both the countries of origin and arrival influence results.

The higher prevalence of stroke in populations from mainland China compared to Western countries[e96] may be due to higher dietary salt intake[e105]; poorly controlled hypertension[e106,107] and less accessible healthcare[e96,108]. Furthermore, the higher stroke prevalence in Black Caribbean's from South London might rely on socioeconomic factors and lifestyle changes that could unmask genetic susceptibilities[e97,109], as Black Caribbean immigrants are more prone to hypertension and diabetes[e110-113].

Similarly, environmental and genetic factors impact the prevalence and morbidity of multiple sclerosis in first-generation immigrants. With migration to higher risk countries, prevalence increases in the migrating population[e91], whereas with migration from high to low risk countries, prevalence decreases[e92]. Genetic factors affect multiple sclerosis rates in the long term, in later generations, notably when massive immigration occurred[e94,114].

Also, it must be borne in mind that the psychological trauma of being displaced, lack of employment, low socioeconomic status, poor housing conditions, social isolation and ethnic discrimination may have limited direct impact on neurological disease; but all these factors can worsen mood disorders and other psychiatric conditions and may prevent access to health care, which may have secondary effects on neurological morbidity and mortality[e115-117].

Current limitations and future directions

Heterogeneity related to study design, exposures, outcome measures, effect modifiers and data presentation limit comparison of study results. In addition, most studies identified in this review were based on retrospective data prone to selection bias. Also, how and where (e.g. outdoor versus indoor) ambient temperatures were measured was not standardized, and the range of temperatures investigated were wide. Studies did not account for factors that may influence temperatures such as humidity, air pollution and geography. In addition, studies comparing migrants to non-migrants were few and did not distinguish between first- and second-generation immigrants; and comparisons were made between migrant populations and either their countries of arrival or departure, but not both.

Importantly, the effect sizes related to global warming and human migration, i.e. exactly how much these factors will influence the morbidity and mortality of neurological disorders, are entirely unknown, as is the influence of other factors associated with climate change such as loss of biodiversity, rising sea levels and drought (**Figure 2**).

Taking the lack of valid data into account, it is not surprising that, so far, predictions of the future epidemiology of neurological disorders are typically based on population growth and life expectancy but completely ignore the impact of climate change and its consequences[e118,119].

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In conclusion, although results were inconsistent due to the heterogeneity of data, our review suggests that climate change will soon change neurological practice because it affects morbidity and mortality of all major neurological disorders. Adequately designed studies to start addressing this issue are urgently needed, which will require coordinated efforts from the entire neurological community.

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TABLE 1	Characteristics of studies included in this review, investigating the association between ambient temperature and major neurological disorders. Temperatures are given as maximum temperatures or increase in mean temperature. Only the effects of high temperatures on epidemiology, symptoms and mortality are listed							
Diagnosis	Number of studies (%)	Study design	Site of study	Total case population	Temperature (maximal)	Effect on epidemiology (Increase in incidence/prevalence/hospitalization)	Worsening of symptoms	Mortality (increase)
Cerebrovascular disorders	61 (70.1%)	Retrospective, observational	AU, CA, CN, DE, DK, ES, GB, IN, IL, IT, JP, KR, PR, QA, RU, SE, TU, TW, US	5,869,284	12.9°C in 1 study 20-27°C in 8 studies >30°C in 17 studies N/A in 2 studies	17 studies	N/A	16 studies
Alzheimer and non-Alzheimer dementia	10 (11.5%)	Retrospective, observational	AU, CN, ES, IT, KR, UK, US, VN	960,675	18°C in 1 study >24.5°C in 1 study >30°C in 8 studies	6 studies	1 study	3 studies
Multiple sclerosis	6 (6.9%)	5 studies: Retrospective, observational 1 study: Retro- and prospective, observational	AU, DE, FR, JP, US	5,305	>20°C	N/A	4 studies	N/A
Headache and Migraine	2 (2.3%)	Retrospective, observational	FR, US	7,156	Temperature increase > 5°C	1 study	1 study	N/A
Parkinson's disease	3 (3,4%)	Retrospective, observational	ES, FR, US	204,656	>31.7°C in 3 studies	N/A	2 study	1 study
Epilepsy	1 (1,1%)	Retrospective, observational	DE	604	>20°C	N/A	Decreased	N/A
Neuroinfectious disorders (TBE)	4 (4.6%)	Retrospective, observational	CZ, SE, SI	N/A	N/A	4	N/A	N/A

Abbreviations: *TBE: Tick-borne encephalitis; Temp: temperature N/A: Not applicable or available;*
AU=Australia, CA=Canada, CN=China, CZ=Czech Republic, DE=Germany, DK=Denmark, ES=Spain, FR=France, GB=United Kingdom, IN=India, IS=Israel, IT=Italy, JP=Japan, KR=South Korea, PR=Puerto Rico, QA=Qatar, RU=Russia, SE=Sweden, SI=Slovenia, TR=Turkey, TW=Taiwan, US=United States, VN=Vietnam

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TABLE 2	Studies (n=8) investigating the differences in neurological disorders between migrants and populations from countries of origin					
Article	Site	Study Design	Recruitment	Population (Participants, Sex, Age)	Country of origin → arrival	Effect on epidemiology (Increase in incidence/prevalence/hospitalization)	Mortality
Cerebrovascular Disease							
[e96]	CN, HK, NAm, SG, TW, WEu,	Observational	Prospective	680 CN, 69.6%M, 65.7±9y; 1,648 HK/ SG/TW, 68%M, 65.5±9.8y; 169 WEu, 71%M, 67.3±9.1y; 441 NAm, 63.3%M, 69.8±10.4y	CN → HK/NAm/SG/TW/WEu	Decreased	No Effect
[e95]	DE, GB, GH, NL	Observational	Retrospective	206 GH, 30,1%M, 52.86±9.9y; 444 DE/GB/NL, 50%M, 52.2±8.8y	GH → GB, DE, NL	Decreased	N/A
[e97]	BB, GB	Observational	Retrospective	665 BB, 42.4%M, 71.2±14.9 271 GB, 66.1±13.7y	BB → GB	Increased	Increased
Multiple Sclerosis							
[e91]	CA, IR	Observational	Retrospective	Onset of MS *before* migration: 29 IR, 20.1%M, 24 (19-31)y Onset of MS *after* migration 48 IR, 33.3%M, 34 (25-40)y	IR → CA	Increased	N/A
[e92]	AU, GB, IE	Observational	Retrospective	331 GB/IE, 208/331 with age 20-49y in 1981	GB, IE → AU	Decreased	N/A
[e93]	Caribbean islands, MQ	Observational	Prospective	53 Afro Caribbean, 13.2%M, 40.7±12.1y 59 MQ, 18.6%M, 43.2±10.4y	Caribbean → MQ	Decreased	N/A
[e94]	BR, IT, ES, PT	Observational	Retrospective	652 BR/ES/IT/PT, 28.4%M, 42y	ES, IT, PT → BR	No Effect	N/A
Dementia							
[e90]	BR, JP	Observational	Prospective	157 JP, 44,6%M, 70-100y	JP → BR	Increased	Increased

Abbreviations: *N/A: Not applicable or available; M=males; MS=multiple sclerosis; y=years;*
AU=Australia, BB=Barbados, BR=Brazil, CN=China, CA=Canada, DE=Germany, ES=Spain, GB=United Kingdom, GH=Ghana, HK=Hong Kong, IE=Ireland, IR=Iran, IT=Italy, JP=Japan, MQ=Martinique NAm=North America, NL=Netherlands, PT=Portugal, SG=Singapore, TW=Taiwan, WEu=Western Europe

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TABLE 3	Studies (n=9) investigating the differences in neurological disorders between migrants and populations from countries of arrival						
Article	Site	Study Design	Recruitment	Population (Participants, Sex, Age)	Country of origin → arrival	Effect on epidemiology (Increase in incidence/prevalence/hospitalization)	Mortality
Cerebrovascular Disease							
e120	IT	Observational, Cross-sectional	Retrospective	3,616.5 migrants (per 100.000), 50.9%M 3,281.1 IT (per 100.000), 50,1%M	Not specified → IT	Increased	N/A
e121	GB	Observational	Prospective	1,540 Caribbean, 25-54y	Caribbean → GB	N/A	Increased
e122	US	Observational	Prospective	746 Hispanic foreign-born, 41.8%M, 64.8y; 15,038 Hispanic and whites US-born, 44.1%M, 65.6y	Hispanic countries → US	Decreased	N/A
e123	NO	Observational	Retrospective	331 SA, 64.7%M; 229 FY, 70%M; 134 SSA, 82.8%M; 124,253 SEA, 0.06%M; 14 CAs, 42.9%M; 15 Cam, 0.4%M; 35-64y for all groups	14 Regions → NO	SA: Increased FY: Increased in men SSA: Increased in men SEA/CAs/CAm: Increased in women	N/A
e124	FI, SE	Observational	Retrospective	2,001 FI, 39%M, 64.7y; 1,574 SE, 36.6%M, 68.4y	FI → SE	N/A	Increased
e125	CN, US	Observational, Cross-sectional	Retrospective	11 CN, 38.1%M, 65.3±13.3 34 US, 43.1%M, 59.8±11.9y	CN → US	Decreased	N/A
Multiple Sclerosis							
e126	AU, GB, IE	Observational	Retrospective	20 GB/IE, 30%M, 32.2y; 99 AU, 37.4%M, 32.6y	GB, IE → AU	No Effect	N/A
e94	BR, IT, ES, PT	Observational	Retrospective	652 BR/ES/IT/PT, 28.4%M, 42y	ES, IT, PT → BR	Increased in IT	N/A
Headache and Migraine							
e127	MX, US	Observational, Cross-sectional	Retrospective	56 Mexican Americans, 23.2%M, >45y; 116 White Americans, 31%M, >45y; 41 Afro-Americans, 19.5%M, >45y	MX → US	No Effect	N/A

Abbreviations: Abbreviations: *N/A: Not applicable or available; M=males; MS=multiple sclerosis; y=years;*
AU=Australia, BR=Brazil, CAm=Central America, CAs=Central Asia, CN=China, ES=Spain, FI=Finland, FY=Former Yugoslavia GB=United Kingdom, IE=Ireland, IT=Italy, MX=Mexico, NO=Norway, PT=Portugal, SA=South Asia, SE=Sverige, SEA=Southeast Asia, SSA=Sub-Saharan Africa, US=United States

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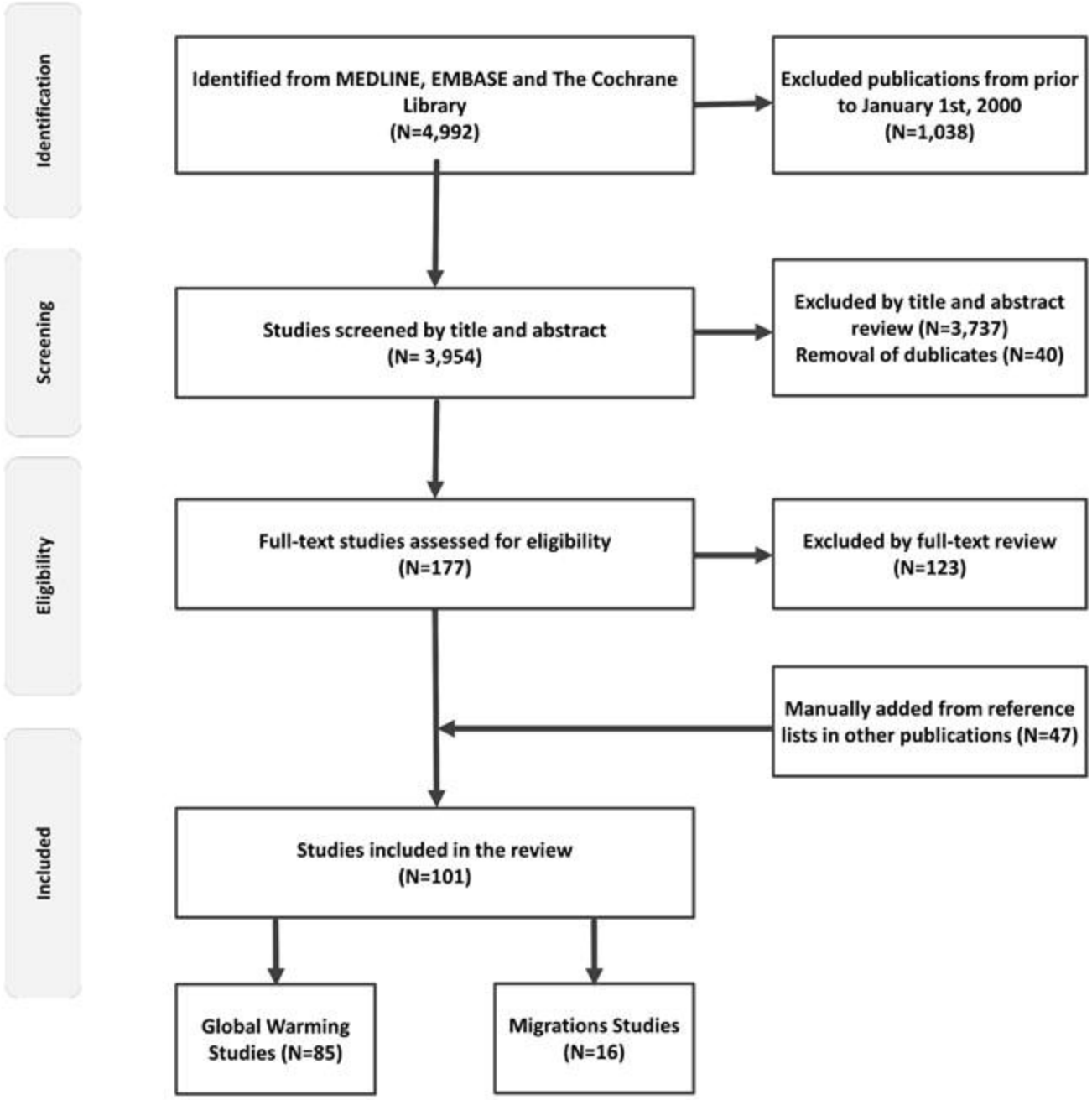
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Figure 1

Flow chart of literature search

Figure 2

Schematic overview on how climate change might soon impact neurological practice. Global warming and human migration were covered in this Review (left). Although we identified no studies addressing precisely how and to what extent rising environmental temperatures may affect neurological disorders and only few studies that investigated neurological disorders in human migrant populations, it seems reasonable to assume that both global warming and climate refugees will alter clinical practice of various neurological disorders owing to alterations in prevalence, incidence, mortality, morbidity and disease semiology. However, global warming and human migration are only two aspects of climate change. Other factors (right) that may change neurological practice directly or indirectly and that were not addressed in this Review include drought, rising sea levels and loss of biodiversity (here, a dead nurse shark), which all might lead to altered neurological practice owing to e.g. food shortage, water insecurity and displacement of communities, as well as an increase in vector-borne diseases (here, a Tsetse fly and an Anopheles mosquito which are the vectors for African trypanosomiasis and cerebral malaria, respectively). *Figure created with Biorender.com*


Identification
Screening
Eligibility
Included
Identified from MEDLINE, EMBASE and The Cochrane Library (N=4,992)
Excluded publications from prior to January 1st, 2000 (N=1,038)
Studies screened by title and abstract (N= 3,954)
Excluded by title and abstract review (N=3,737)
Removal of dublicates (N=40)
Full-text studies assessed for eligibility (N=177)
Excluded by full-text review (N=123)
Manually added from reference lists in other publications (N=47)
Studies included in the review (N=101)
Global Warming Studies (N=85)
Migrations Studies (N=16)

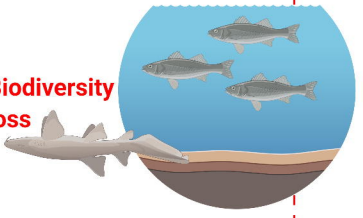


Vector borne diseases



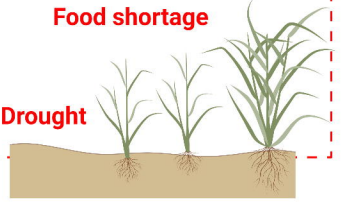
Rising sea levels

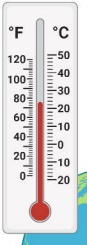
Biodiversity loss

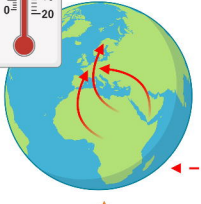


Food shortage

Drought

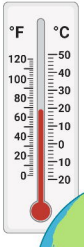


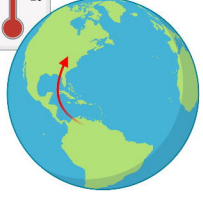




Global warming

Alzheimer's dementia: Agitation, mortality, hospital admissions
Epilepsy: Hospital admissions, seizure frequency
Headache: Migraine triggers
Multiple sclerosis: Symptoms, cognition
Parkinson's disease: Autonomic failure
Stroke: Hospital admissions, mortality
TBE: Geography





Alzheimer's dementia: Prevalence
Multiple sclerosis: Prevalence, symptoms
Stroke: Prevalence

Human migration